

For an arithmetic series:

$$u_n = a + (n-1)d, \quad S_n = \frac{1}{2}n(a+l) = \frac{1}{2}n\{2a + (n-1)d\}$$

For a geometric series:

$$u_n = ar^{n-1}, \quad S_n = \frac{a(1-r^n)}{1-r} \quad (r \neq 1), \quad S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

Binomial series:

$$(a+b)^n = a^n + \binom{n}{1} a^{n-1}b + \binom{n}{2} a^{n-2}b^2 + \binom{n}{3} a^{n-3}b^3 + \dots + b^n, \text{ where } n \text{ is a positive integer}$$

$$\text{and } \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots, \text{ where } n \text{ is rational and } |x| < 1$$

# Trigonometric Cheat Sheet

## 0. Pythagorean Identities

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

## 1. Sum and Difference

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

## 2. Double Angle

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$$

$$= 2 \cos^2 \theta - 1$$

$$= 1 - 2 \sin^2 \theta$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

## 3. Product-to-Sum

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

## 4. Sum-to-Product

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

## 5. Power-Reducing

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4}$$

$$\cos^3 \theta = \frac{3 \cos \theta + \cos 3\theta}{4}$$

$$\sin^4 \theta = \frac{3 - 4 \cos 2\theta + \cos 4\theta}{8}$$

$$\cos^4 \theta = \frac{3 + 4 \cos 2\theta + \cos 4\theta}{8}$$

$$\sin^5 \theta = \frac{10 \sin \theta - 5 \sin 3\theta + \sin 5\theta}{16}$$

$$\cos^5 \theta = \frac{10 \cos \theta + 5 \cos 3\theta + \cos 5\theta}{16}$$

## 6. Harmonic Form

If  $a$  and  $b$  are positive:

$$a \sin x + b \cos x = R \sin(x + \alpha)$$

$$a \sin x - b \cos x = R \sin(x - \alpha)$$

$$a \cos x + b \sin x = R \cos(x - \alpha)$$

$$a \cos x - b \sin x = R \cos(x + \alpha)$$

$$R = \sqrt{a^2 + b^2}, \quad R \cos \alpha = a, \quad R \sin \alpha = b$$

$$\cos^2 \theta + \sin^2 \theta \equiv 1,$$

$$1 + \tan^2 \theta \equiv \sec^2 \theta,$$

$$\cot^2 \theta + 1 \equiv \operatorname{cosec}^2 \theta$$

$$\sin(A \pm B) \equiv \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) \equiv \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) \equiv \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\cos 2A \equiv \cos^2 A - \sin^2 A \equiv 2 \cos^2 A - 1 \equiv 1 - 2 \sin^2 A$$

$$\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

## Half Angle Formulas

$$\sin \theta = \pm \sqrt{\frac{1 - \cos(2\theta)}{2}}$$

$$\cos \theta = \pm \sqrt{\frac{1 + \cos(2\theta)}{2}}$$

$$\tan \theta = \pm \sqrt{\frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}}$$

## Sum and Difference Formulas

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

## Product to Sum Formulas

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$$

## Sum to Product Formulas

$$\sin \alpha + \sin \beta = 2 \sin \left( \frac{\alpha + \beta}{2} \right) \cos \left( \frac{\alpha - \beta}{2} \right)$$

$$\sin \alpha - \sin \beta = 2 \cos \left( \frac{\alpha + \beta}{2} \right) \sin \left( \frac{\alpha - \beta}{2} \right)$$

$$\cos \alpha + \cos \beta = 2 \cos \left( \frac{\alpha + \beta}{2} \right) \cos \left( \frac{\alpha - \beta}{2} \right)$$

$$\cos \alpha - \cos \beta = -2 \sin \left( \frac{\alpha + \beta}{2} \right) \sin \left( \frac{\alpha - \beta}{2} \right)$$

# POWER REDUCING FORMULA

## SQUARES

$$\sin^2 u = \frac{1 - \cos 2u}{2}$$

$$\cos^2 u = \frac{1 + \cos 2u}{2}$$

$$\tan^2 u = \frac{1 - \cos 2u}{1 + \cos 2u}$$

## FOURTHS

$$\sin^4 u = \frac{3 - 4 \cos 2u + \cos 4u}{8}$$

$$\cos^4 u = \frac{3 + 4 \cos 2u + \cos 4u}{8}$$

$$\tan^4 u = \frac{3 - 4 \cos 2u + \cos 4u}{3 + 4 \cos 2u + \cos 4u}$$

## CUBES

$$\sin^3 u = \frac{3 \sin u - \sin 3u}{4}$$

$$\cos^3 u = \frac{3 \cos u + \cos 3u}{4}$$

$$\tan^3 u = \frac{3 \sin u - \sin 3u}{3 \cos u + \cos 3u}$$

## FIFTHS

$$\sin^5 u = \frac{10 \sin u - 5 \sin 3u + \sin 5u}{16}$$

$$\cos^5 u = \frac{10 \cos u + 5 \cos 3u + \cos 5u}{16}$$

$$\tan^5 u = \frac{10 \sin u - 5 \sin 3u + \sin 5u}{10 \cos u + 5 \cos 3u + \cos 5u}$$



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POWER-REDUCING FORMULAS

If **a** and **b** are positive

*a sin x + b cos x can be written in the form R sin(x + α)*

*a sin x - b cos x can be written in the form R sin(x - α)*

*a cos x + b sin x can be written in the form R cos(x - α)*

*a cos x - b sin x can be written in the form R cos(x + α)*

$$R = \sqrt{a^2 + b^2}$$

$$R \cos \alpha = a \text{ and } R \sin \alpha = b$$

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$